



WINA SORTA MARIA GEA

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North Sumatra, Indonesia

A final-year Computer Technology student at Del Institute of Technology with strong interests in Web Development, Front-End Engineering, UI Design, and IoT. Highly curious and enthusiastic about technological innovation, meticulous in software testing, and committed to continuous learning. Responsible, adaptable, and eager to contribute to building reliable and sustainable technology within a collaborative team environment.

Education

Institut Teknologi Del

Aug 2023 - Sep 2026 (Expected)

Diploma in Computer Technology, 3.21/4.00

- Relevant Courses: Web Development and Internet Application Development, Software Engineering (Design, Analysis, and Quality Assurance), Network Security and Cryptography, Programming (C, JavaScript, Python, PHP, Laravel, HTML/CSS), Algorithms and Data Structures, Database Systems, Operating Systems, Embedded Systems Fundamentals, IoT Technology, Cloud Computing, Computer Networks, Discrete Mathematics, Linear Algebra, Probability and Statistics, Computer Architecture, and Digital Innovation.

Projects

Computer Technology Career Information Website

Jan 2024 - Jun 2024

Project Leader

This website functions as a career information platform for Computer Technology students to obtain job information, career guidance, and self-development according to the three main cores of Network Computing, Cloud Computing, and Embedded Systems.

- Engineer and implement a dynamic web platform dedicated to providing comprehensive career insight and job type information for Computer Technology graduates, which directly addresses users' information needs.
- Developed full system functionality, including alumni tracking, job listings, and career path filtering, using WordPress and a robust framework to ensure system scalability and efficient data handling.
- Manage and structure MySQL databases to support complex relational data for job information, optimizing query performance for fast information retrieval.

IoT-Based Air Quality Monitoring System

Jan 2025 - Jun 2025

Project Leader

Led the development of an IoT-based air quality monitoring system using ESP32, DHT22, SGP30, and PM2.5 sensors to capture real-time indoor air data and send it to the Blynk Cloud platform.

Designed the user interface (UI) for the I2C LCD screen and Blynk dashboard, ensuring clear visualization of temperature, humidity, VOC, eCO2 and PM2.5 levels.

Prepare complete project documentation, including system architecture, hardware configuration, data flow design, feature

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- Prepare complete project documentation, including system architecture, hardware configuration, data flow design, feature descriptions, and testing procedures

IoT-Based Smart Walking Stick with Position and Posture Monitoring for The Visual Impaired

Aug 2025 - Jan 2025

Project Leader

An IoT assistive technology project integrating smart walking stick sensors, GPS–4G tracking, and IMU-based posture monitoring to support real-time navigation and safety for visually impaired users.

- Led the development of a dual-system IoT solution, consisting of a smart walking stick built with an Arduino Mega (ultrasonic, water and light sensors) and an ESP32 module with 4G and GPS, fully integrated into a custom-designed PCB.
- Designed a user interface for visualization in Unreal Engine and built a Python-based GUI to present real-time posture monitoring data from MPU sensor
- Prepared comprehensive project documentation, including system architecture, hardware integration workflows, communication protocols, posture classification design, and testing procedures.

Organisational Experience

Department of Communication and Information - Institut Teknologi Del

Oct 2024 - Present

Creative Division Member

- Support the Student Executive Board's internal and external communications through the creation of visual content and media publications.
- Providing creative ideas and suggestions for content development and improving design quality.
- Create graphic designs and digital content for posters, feeds and other promotional materials.
- Collaborate with other departments to ensure effective information dissemination and consistent organizational branding.

Skills

- **Programming:** C, Java, Python, PHP, JavaScript, SQL, HTML, CSS, Laravel, Next.js
- **Soft Skills:** Strong teamwork and collaboration abilities, effective communication skills, excellent time management, creative and analytical thinking, and proven leadership experience.
- **Tools:** GitHub, Visual Studio Code, Postman, WordPress, Arduino IDE, XAMPP, phpMyAdmin, Node-RED, Docker Desktop, Kubernetes
- **Language:** Bahasa Indonesia (Native), English (Intermediate)